Harish Manoharan

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Software engineer at Purgo AI working on agents, retrieval and evals. M.S. Computer Science, UT Dallas, 2026.

EXPERIENCE

Purgo AI San Francisco Bay Area
AI startup: an agent that turns data-engineering tickets into PySpark, SQL and dbt code
Software Engineer Jun 2026 – Present
Software Engineer Intern (on site, then remote during M.S.) Jun 2025 – May 2026

- Cut missed source tables from 79 to 24 on our 103-ticket internal benchmark by adding dbt lineage expansion (SQL parsing with sqlglot) to the LangGraph agent's retrieval, with no ticket losing a table it had already found.
- Built the Polaris/Iceberg catalog search behind the agent's retrieval (Pinecone, keyword fallback): recall@10 0.68 vs 0.54 and 0.18 s vs 4.0 s against Databricks Vector Search on 188 queries, plus PII/PHI masking (Presidio) on sampled rows.
- Turned recurring Databricks runtime failures into AST lint rules that the agent's self-review runs on generated PySpark/SQL, cutting runtime failures, plus a recovery pass that keeps the best-scoring candidate when all fail.
- Rewrote the retrieval and drafting prompts for a smaller reasoning model and set reasoning effort per stage, switching only after 5 runs on 102 tickets showed retrieval recall held within noise. Those stages now cost 7.6x less.
- Built the compliance-validation (IQ/OQ) engine that checks life-sciences customers' Databricks platforms on Azure and AWS (SigV4): it runs a git-versioned library of 59 tests as cancellable jobs with per-test timeouts, keeps redacted evidence for signed audit reports, and turns requirement documents into new tests with an LLM.
- Built the LLM validation product across UI, API and test engine (React/Next.js, NestJS, PostgreSQL): 10 behavioral tests on Azure OpenAI or Databricks-served models (hallucination on missing fields, extraction F1, demographic bias, 20-run determinism, latency at 20 concurrent calls) and 9 checks on an Azure deployment's access, network and audit setup.

Digitus Business Solutions Bengaluru, India
Software Engineer Intern May 2024 – Jul 2024

- Raised SQL execution accuracy from 52% to 76% on a 500-query evaluation set by building the text-to-SQL agent behind a database chatbot used mainly by the sales team, with schema discovery and retries driven by execution errors.
- Built the integration that submits invoices from QAD ERP to Malaysia's MyInvois tax system: each document digitally signed and sent base64-encoded, with validation results and document events tracked for clients.

PwC India Bengaluru, India
Technology Consulting Intern, Data Analytics May 2023 – Jul 2023

- Built a customer analytics platform served through Django REST APIs and Power BI, with XGBoost + SARIMA demand forecasts that cut forecast error (MAPE) 15% against the baseline across 50+ SKUs.

PROJECTS

- **ReadableSTEM** (readablestem.com, open source, live): turns STEM course PDFs into accessible web pages with MathML equations. An LLM pipeline (Claude, FastAPI) checks each equation against the PDF and flags uncertain ones for instructor review. Heading F1 **0.96** on a pre-registered held-out set.
- **Margins** (studymargins.com, open source, live): turns a student's course files into cited notes, quizzes and practice exams. Hybrid keyword and vector retrieval (Claude, pgvector) finds **99%** of relevant passages on a 38-topic eval.

EDUCATION

The University of Texas at Dallas M.S. Computer Science GPA 3.9/4.0 · Dean's Graduate Scholar May 2026
Indian Institute of Technology Madras Dual Degree: M.Tech Data Science, B.S. Biological Sciences May 2024
INSPIRE Scholar · Young Research Fellow · Thesis with The Jackson Laboratory on unsupervised behavior discovery in mice

SKILLS

LLMs: RAG, evals, AI agents (LangGraph), PyTorch · **Cloud and tools:** AWS, Azure, GCP, Docker, Git, GitHub Actions
Data: Databricks, Apache Spark (PySpark), dbt, SQL query parsing (sqlglot), PostgreSQL, Apache Iceberg, Pinecone
Languages and web: Python, TypeScript, JavaScript, SQL, FastAPI, NestJS, REST APIs, React, Next.js